

Experiment No-11

Waveform Synthesis Using Laplace transforms

AIM: Finding the Laplace transform & Inverse Laplace transform of some signals.

Software Required: Matlab software

Theory:

Bilateral Laplace transforms:

The Laplace transform of a signal $f(t)$ can be defined as follows:

$$F(s) = \mathcal{L}\{f(t)\} = \int_{-\infty}^{\infty} e^{-st} f(t) dt.$$

Inverse Laplace transform

The inverse Laplace transform is given by the following formula :

Program:

```
clc;
clear all;
close all;
%representation of symbolic variables
syms f t w s;
%laplace transform of t
f=t;
z=laplace(f);
disp('the laplace transform of f = ');
disp(z);

% laplace transform of a signal
%f1=sin(w*t);
f1=-1.25+3.5*t*exp(-2*t)+1.25*exp(-2*t);
v=laplace(f1);
disp('the laplace transform of f1 = ');
disp(v);
lv=simplify(v);
```

```

pretty(lv)
%inverse laplace transform
y1=ilaplace(z);
disp('the inverse laplace transform of z = ');
disp(y1);
y2=ilaplace(v);
disp('the inverse laplace transform of v = ');
disp(y2);
ezplot(y1);
figure;
ezplot(y2)

```

Output:

the laplace transform of f =
 $1/s^2$

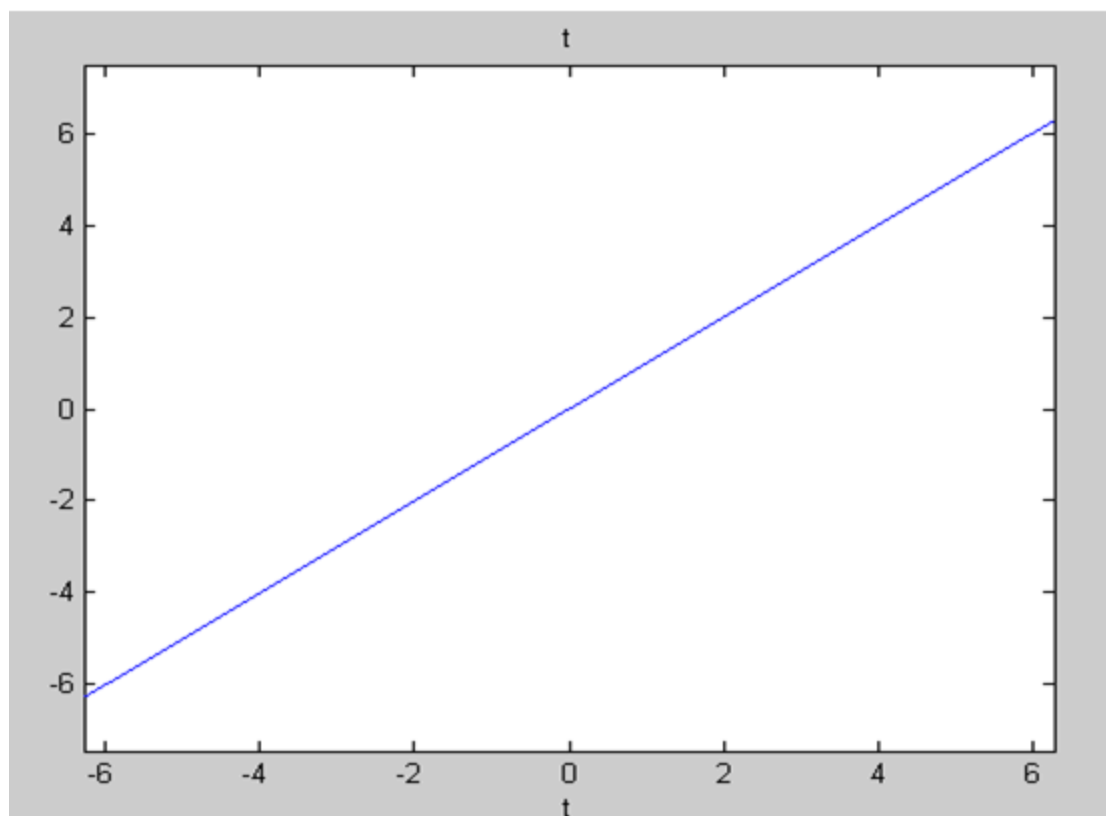
the laplace transform of f1 =
 $5/(4*(s + 2)) + 7/(2*(s + 2)^2) - 5/(4*s)s$

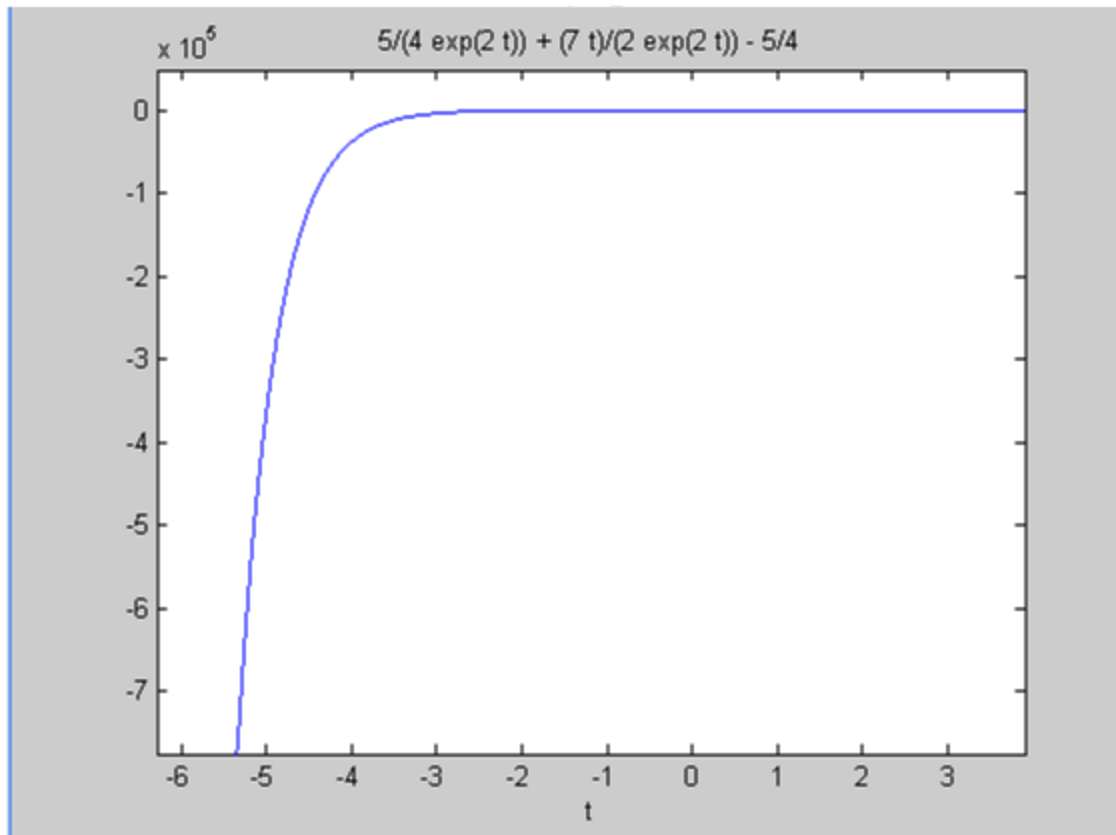
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 $s (s + 2)^2$

the inverse laplace transform of z =t

the inverse laplace transform of v =
 $5/(4*\exp(2*t)) + (7*t)/(2*\exp(2*t)) - 5/4$



**VIVA QUESTIONS:-**

1. Define Laplace-Transform?
2. What is the Condition for Convergence of the L.T?
3. What is the Region of Convergence (ROC)?
4. State the Shifting property of L.T?
5. State convolution Property of L.T?